

TechPunjab (SIH 1629)

Complete Architecture, Concept Workflows & Technical Reference Notes

Project Name: TechPunjab (GigMint)	Target Track: SIH 1629
Government Body: Punjab Skill Development Mission (PSDM)	Tech Stack: Next.js 14, TypeScript, Tailwind, Gemini AI
Core Protocols: UPI Milestone Escrow & Merkle Ledger	Design System: Bento Studio (Theme 3)

DOCUMENT OBJECTIVE (■■■■■■■■):

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1. Problem Statement & High-Level Context (SIH 1629)

1.1 Problem Statement (The Real-World Gap):
The current ecosystem for MSMEs in Punjab is fragmented and inefficient—spanning sectors like (Auto Parts & Textiles), (Sports Goods & Leather), (Agro-Food Processing), and (IT & Startups)—where MSMEs face significant challenges in scaling and accessing resources. Key pain points include:
• High Agency Cost & Distance: MSMEs incur high costs and travel distances to interact with government agencies.
• Trust Deficit (Lack of Digital Identity): MSMEs lack a verifiable digital identity, leading to mistrust and delayed transactions.
• Fake Credentials & Unverified Resumes: MSMEs face issues with fake credentials and unverified resumes, impacting their credibility.
• Unstructured Project Scoping: MSMEs struggle with unstructured project scoping, leading to inefficiencies and disputes.

1.2 Solution Pillars: TechPunjab Ecosystem (The Solution Pillars):
The solution is built on four pillars, leveraging the Punjab Skill Development Mission (PSDM) and the National Skill Qualification Framework (NSQF) levels 4, 5, and 6. These pillars are:
1. Native UPI Smart Escrow
2. Cryptographic PSDM Ledger
3. Gemini AI Milestone Scoper
4. MSME Digital Identity (Pillars)

Pillar	Technical Working	MSME & Trainee Impact
1. Native UPI Smart Escrow	Enables secure transactions between MSMEs and buyers/sellers using UPI VPA, ensuring funds are held in escrow until the transaction is complete.	0% risk of fraud or payment disputes, ensuring secure transactions and faster delivery.
2. Cryptographic PSDM Ledger	Utilizes SHA-256 cryptographic hashing to create a tamper-proof ledger of transactions and credentials, ensuring data integrity.	100% secure and verifiable transactions, ensuring trust and transparency.
3. Gemini AI Milestone Scoper	Uses Gemini AI to analyze project requirements and generate structured milestones (Database, Core Logic, Production) for MSMEs.	MSMEs receive clear, structured project milestones and guidance, improving project management and delivery.

4. Kanban Contract Workspace	■■■■■■ ■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■■■■■ ■■ ■■■■■■■■ ■■■■ ■■■■■■ 'Completed' ■■■■ ■■ ■■■■■■■■■■ ■■■■■■ ■■■■■■ ■■■■ ■■■■	■■■■-■■■■ ■■■■■■■■ ■■■■■■■■■■ ■■ ■■■■■■■■ ■■■■-■■■■■■■ ■■■■■■■■■■
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2. System Architecture & Technology Stack

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3. Detailed Concept & Workflow Breakdown (100% Complete)

3.1 Dual-Persona Architecture (■■■■■■■■■■ vs ■■■■■■■■ ■■■■■■■■■■)

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■■■■■■■■ ■■■■■■■■: context/AppContext.tsx, components/Navbar.tsx, app/page.tsx (Floating Dock)■■

■■■■■■■■ ■■■■■■■■■■ (Step-by-Step Flow):

- [illegible]

3.2 Entrance Splash Screen & Welcome Onboarding (StartingSplash & WelcomeModal)

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■■■■■■■■ ■■■■■■: components/StartingSplash.tsx, components/WelcomeModal.tsx■■

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- StartingSplash (z-index: 100) 3.1 'Loading Platform...'
- 'Skip intro'
- WelcomeModal 3 (UPI Escrow, PSDM Ledger, AI Scoper) (Trainee MSME)

3.3 Bento Studio Dashboard (BentoGrid.tsx)

[illegible]

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- [illegible]

3.4 Native UPI Milestone Escrow Protocol (EscrowSimulator.tsx) - THE CORE USP!

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■■■■■■■■■■ ■■ ■■■■■■■■■■ MSME ■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■ 100% ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■

4-■■■■■ ■■■■■ ■■■■■■■■■■ (4-Step Complete Lifecycle):

Step (Step)	Escrow State	Action & Logic
Step 1: Client Fund Lock	ESCROW_LOCKED	Escrow amount is locked (₹15,000) and UPI QR (VPA: psdm.escrow@icici) is generated. Funds are locked and the QR is shared with the client.
Step 2: Trainee Work Submission	WORK_SUBMITTED	Trainee submits work (.zip file, GitHub PR link, Swagger API spec) and the status is updated to IN_REVIEW.
Step 3: Client Approval & UPI Release	DISBURSED_TO_FREELANCER	Client approves and disburses funds. UPI VPA (gurpreet@okhdfcbank) is used for release. Funds are released to the freelancer.
Step 4: Revision / Dispute Fallback	REVISION_REQUESTED	Client requests revision or dispute. If revision is requested, the status is updated to REVISION_PENDING. If a dispute is raised, the status is updated to DISPUTE_PENDING.

3.5 AI Project Scope & Bid Generator (AIScoperModal.tsx)

Project Scope: Google Gemini 1.5 Flash **Project Details:** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover

Capabilities: (Dual-Mode Capabilities):

- Mode A (Client Scoper):** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover **Project Details:** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover **Project Details:** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover
- Mode B (Freelancer Pitch Generator):** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover **Project Details:** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover

3.6 PSDM On-Chain Skill Ledger (SkillLedgerModal.tsx)

Project Scope: Google Gemini 1.5 Flash **Project Details:** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover

Capabilities:

- Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover
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3.7 Contract Sprint Kanban Workspace (KanbanWorkspace.tsx)

Project Scope: Google Gemini 1.5 Flash **Project Details:** MSME **Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover

Capabilities: (Core Rules):

- Project ID:** 3 **Project Name:** AI Forecasting Model & Real-time Stock Alerts (WhatsApp>Email hooks) **Project Description:** Executive Analytics Dashboard, RBAC Auth & Production Deployment Handover
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3.8 Punjab MSME Marketplace & Quick Bid (ExploreGigs.tsx & QuickBidDrawer.tsx)

Functionalities: 1-MSME AI-PSDM

- Categories: (Agro-Food), (IT & Silicon Belt), (Auto & Engineering), (Sports & Leather)
- Skills: Web & AI, IoT & Automation, Cloud & Data
- AI Quick Bid Drawer: 'Quick Bid' MSME -PSDM AI MSME Udyam / GSTIN PSDM

3.9 Authentication & Government SSO (LoginModal.tsx & /login/page.tsx)

Functionalities: MSME-PSDM (SSO)

- 1-Click Demo Login: MSME (NSQF L5) MSME Employer
- PSDM (PB-PSDM-2024-AI-89421) MSME Udyam / GSTIN
- OTP: 4-OTP (8942)

4. End-to-End User Journeys (■■■■-■■-■■■ ■■■■ ■■■■■■)

Scenario A: ■■ MSME ■■■■■■ (■■■■■■ ■■■■■■, ■■■■■■ ■■■■■■■■■■■■) ■■ ■■■■■■:

1. **[REDACTED]: [REDACTED] [REDACTED]** [REDACTED]
[REDACTED]
2. **AI Scoper** [REDACTED]: [REDACTED] 'AI Scoper' [REDACTED] [REDACTED] [REDACTED] 'Build an automated inventory tracker for retail'[REDACTED]
AI [REDACTED] 3 [REDACTED] ([REDACTED]5,000, [REDACTED]10,000, [REDACTED]5,000) [REDACTED]
3. **[REDACTED] [REDACTED]: [REDACTED]** [REDACTED]
[REDACTED] [REDACTED] [REDACTED] 5,000 UPI [REDACTED]
4. **[REDACTED] [REDACTED]: [REDACTED]** [REDACTED]
[REDACTED] [REDACTED] 'IN_REVIEW' [REDACTED]
5. **[REDACTED] [REDACTED]: [REDACTED]** [REDACTED] [REDACTED] [REDACTED]
UPI [REDACTED] 5,000 [REDACTED], [REDACTED]

Scenario B: ■■ ■■■■ ■■■■ (■■■■■■■■ ■■■■, MSDC ■■■■■■ ■■■■■■) ■■ ■■■■■■:

1. **NSQF L5 Assessment:** PSCs will assess candidates based on their skills and knowledge. The assessment will be conducted in a computer-based format. **NSQF L5** assessment will be conducted on **10th April**.
2. **AI Interview:** AI-based interview will be conducted on **11th April**. The AI will assess the candidates based on their responses to the questions. **98%** of the candidates will be shortlisted for the next round.
3. **'Quick Bid' AI Interview:** AI-based interview will be conducted on **12th April**. The AI will assess the candidates based on their responses to the questions. **10%** of the candidates will be shortlisted for the next round.
4. **'In Progress' AI Interview:** AI-based interview will be conducted on **13th April**. The AI will assess the candidates based on their responses to the questions. **10%** of the candidates will be shortlisted for the next round.
5. **Final Interview:** Final interview will be conducted on **14th April**. The final interview will be conducted in a computer-based format. **10%** of the candidates will be shortlisted for the next round. **3.84,000** candidates will be shortlisted for the next round.

5. SIH Evaluator & Viva Q&A Cheat Sheet (100% ✓ 100% ✓ 100% ✓ 100% ✓ 100% ✓)

(b) (5) SIH [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED], [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED].

[illegible]

Q2: [REDACTED] [REDACTED] [REDACTED] [REDACTED] (Dispute) [REDACTED] [REDACTED] [REDACTED]?	[REDACTED] [REDACTED] REVISION_REQUESTED [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] MSME [REDACTED] [REDACTED], [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]; [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
Q3: AI Scoper [REDACTED] [REDACTED] 3 [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]?	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] (Agile Methodology) [REDACTED] [REDACTED]: 1) Architecture & Data Schema, 2) Core Business Logic & AI Engine, 3) Production UI & Handover[REDACTED] [REDACTED] 3-[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
Q4: [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]?	[REDACTED] Next.js 14 [REDACTED] [REDACTED] [REDACTED], RESTful/tRPC [REDACTED], [REDACTED] [REDACTED]/[REDACTED] [REDACTED] [REDACTED] [REDACTED] SHA-256 [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]